Code No: 136FD **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year II Semester Examinations, February - 2023
GEOINFORMATICS
(Common to EEE, CSE, MIE)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

JNTUH Used papers 2023

PART - A

(25 Marks)

1.a)	What is meant by Focal length of lens in aerial photography?	[2]
b)	Why is stereo photogrammetry important?	[3]
c)	Write down principle of remote sensing.	[2]
d)	What is sensor in geology?	[3]
e)	What do you understand by image histogram?	[2]
f)	Is image classification supervised or unsupervised?	[3]
g)	List few urban applications of GIS.	[2]
h)	What is the definition of error in GIS?	[3]
i)	How do you explain GNSS?	[2]
j)	Which countries use GLONASS?	[3]

PART - B

(50 Marks)

2.a) Discuss about various types of Aerial Photogrammetric techniques.
b) What are the factors effecting aerial photographs? [5+5]

OR

3.a) What are the advantages of digital photogrammetry?
b) How can you interpret the aerial photographs? Discuss. [5+5]

4. List and explain any two satellites and write down characteristics of satellites. [10]

OR

5. What are Spectral reflectance curves? Discuss them for water, vegetation and borewell. [10]

6.a) What are the main characteristics of satellite imageries? Explain.
b) Discuss the file formats used in various satellites. [5+5]

OR

7. Explain the applications of remote sensing in ground water prospects and potential recharge zones. [10]

8. What are the data types in GIS? Explain with clear illustrations the concepts of various data models used in GIS. [10]

OR

9.a) Discuss in brief various applications of GIS in civil engineering.

b) What do you understand by spatial data and how are they integrated to make a GIS? [5+5]

10.a) Explain in brief how GPS helps in modern life.

b) Discuss in brief about GPS satellite signals and also explain its importance. [5+5]

OR

11.a) Discuss about composition of GPS satellite signals.

b) What is differential GPS used for? Discuss the principle of differential GPS. [5+5]

---ooOoo---

JNTUH Used papers 2023